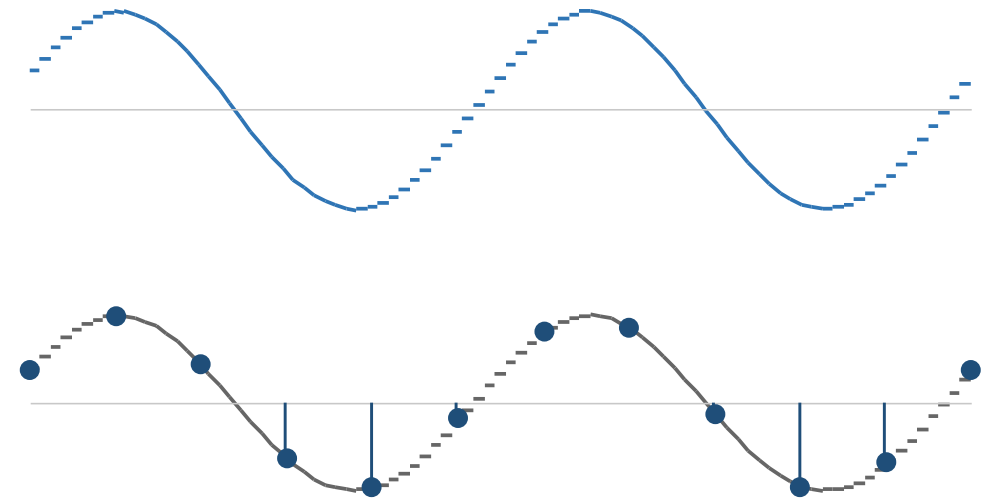


Topic 1

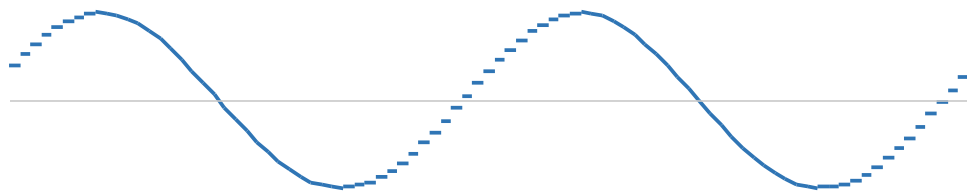
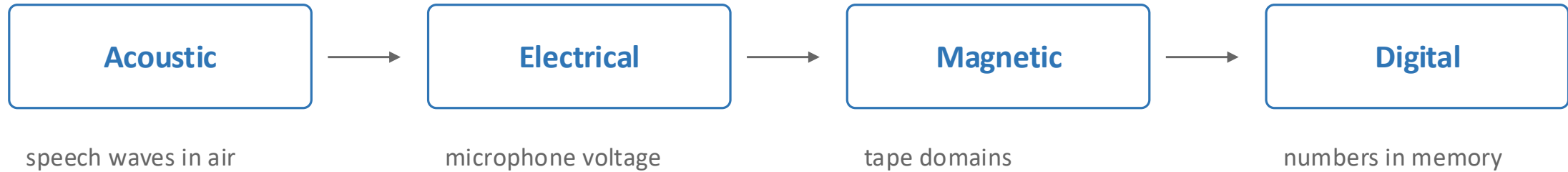
Introduction to Signals and Systems

Key idea. A signal carries information through a physical pattern; a system transforms one signal into another.

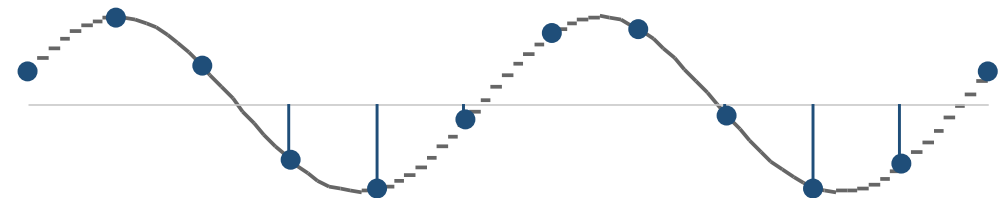


Signals: information as physical variation

A signal is a pattern of variation in a physical quantity. The same information can move through several equivalent representations.



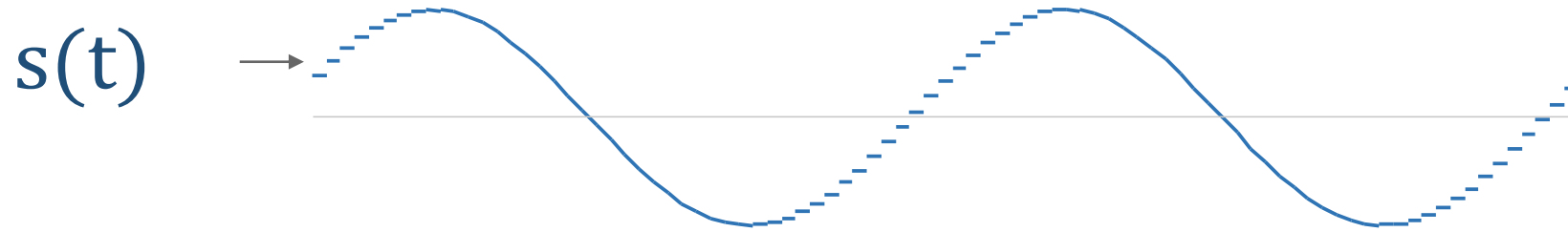
continuous physical pattern



discrete numerical sequence

Mathematical representation: continuous time

A continuous-time signal is defined for every value of a continuous independent variable, usually time.



Examples

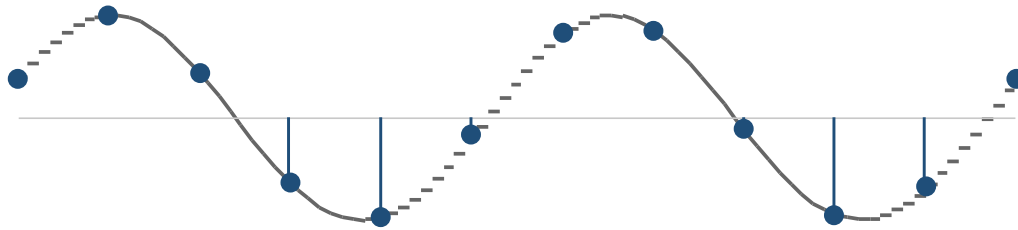
- voice pressure as a function of time
- image intensity as a function of position
- temperature as a function of time and place

$$s(x,y)$$

For images, the independent variables are spatial coordinates instead of time.

Discrete-time signals: sampling a continuous signal

Digital processing requires a sequence of values. Sampling takes snapshots of a continuous-time signal at equally spaced instants.



$$s[n] = s(nT_s)$$

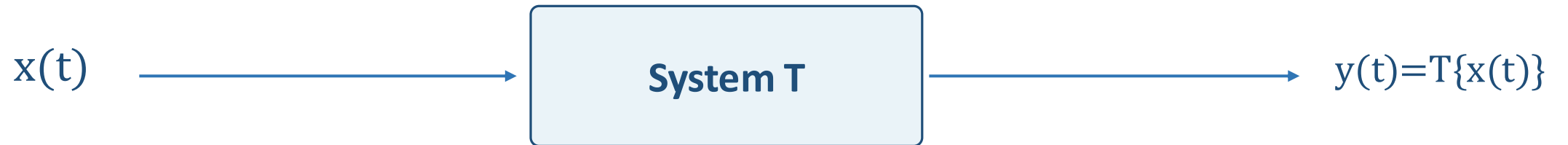
n is an integer sample index

T_s is the sampling period



Systems: transformations of signals

A system is an operator or process that transforms an input signal into an output signal.



Example: squarer

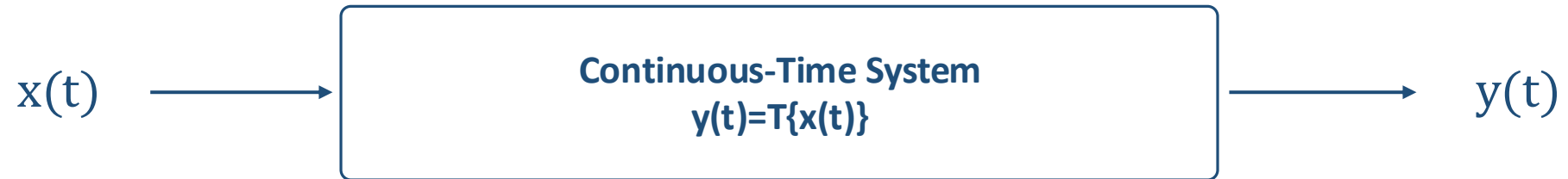
$$y(t)=x^2(t)$$

$$y[n]=x^2[n]$$

Same idea, continuous or discrete.

Block diagrams: visual language for systems

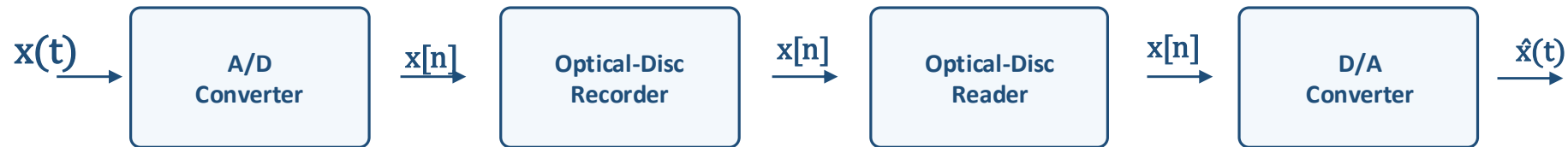
Engineers use block diagrams to show operations, signal paths, and processing sequences.



Key idea. The diagram does not replace the equation; it clarifies where each operation happens and how signals flow.

Example: audio CD recording and playback

A complete audio system connects physical sound, digital storage, and reconstruction.



What changes at each stage?

- A/D converts continuous voltage to numbers
- Recorder and reader store/retrieve the sequence
- D/A reconstructs a continuous-time output $\hat{x}(t)$

Continuous vs. discrete domains

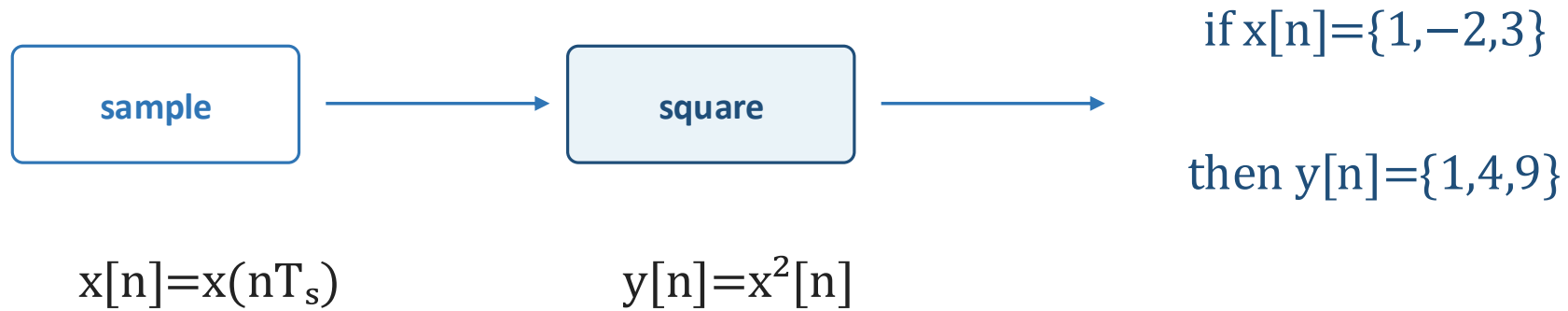
The same ideas appear in two domains, but the independent variable and notation change.

Property / Operation	Continuous Time	Discrete Time
Independent variable	$t \in \mathbb{R}$	$n \in \mathbb{Z}$
Signal form	$s(t), x(t)$	$s[n] = s(nT_s), x[n]$
System mapping	$y(t) = T\{x(t)\}$	$y[n] = T\{x[n]\}$
Example: squarer	$y(t) = x^2(t)$	$y[n] = x^2[n]$

Key idea. Always check which domain you are working in before interpreting an equation or block diagram.

Mini-example: sampling and squaring

Suppose we sample a continuous signal and then apply a squarer system.



The system acts sample by sample because this example has no memory.

What you should recognize after this topic

By the end of this introduction, you should be comfortable moving between words, diagrams, and equations.

Signal

information in a varying quantity

Continuous-time

defined for every t

Discrete-time

indexed by integer n

System

operator $T\{\cdot\}$

Block diagram

visual processing flow

Audio pipeline

A/D $\rightarrow$ store $\rightarrow$ D/A

Notation

Symbol	Meaning
t	Continuous time variable (independent variable)
n	Discrete sample index (integer, $n \in \mathbb{Z}$)
T_s	Sampling period (time interval between consecutive samples)
$s(t), x(t)$	Continuous-time input signals
$s[n], x[n]$	Sampled discrete-time signals / sequences
$y(t), y[n]$	Continuous-time and discrete-time system output signals
$\mathcal{T}\{\cdot\}$	System transformation operator
$\hat{x}(t)$	Reconstructed continuous-time output signal